



Western University Faculty of Engineering

Lab Section 14, Jonathon Southen

Team Identifier: A

Team Members: Leo Bogaert, Karim Estafan,

Sarvesh Mohan Kumar, Matthew Ji,

Alexander Cecco, Mohammed Amjad

Design Documentation: Hospital Tracking

Design Description: Bluetooth based asset tracking system

1 Final Design Documentation

1.1 Description

Our design has 2 parts the hardware component and the software component.

Hardware:

Our trackers are generic mass-produced BLE modules powered by CR2032 coin cell batteries. Each tracker will be placed into a 3D-printed case and zip-tied to the asset to be tracked. Each tracker will have a projected battery life of ~2 years and will send out an advertisement every ~10 seconds. In each room of the hospital, an existing Cisco Access point will receive this signal. Rooms without existing infrastructure will use an ESP32 to receive this signal. These are called nodes, and by using a Log-Distance Path Loss Mode, given by: $d = 10^{\frac{A-RSSI}{10n}}$, where d is the distance, A is the signal strength at 1 meter, RSSI is the signal strength and n is an environmental constant, we can calculate the distance of each tracker to each node. These calculations will then be sent to an MQTT server run using the eclipse mosquito software.

Software:

The website is built on TypeScript, JavaScript, and React. It displays the best location of each asset on a floor plan map of the building, based on multiple past observations of the tracker's location. It gets its information from the MQTT broker using a paho client, which connect and updates the data. This data is stored in local async storage to be used later. Once data is received, the assets are each assigned a room based on the node with the closest distance reading. Assets can be renamed and assigned tags for better organization. The assets can be searched for using room names, asset names, and usage tags for easy identification of location.

1.2 Design Drawings

1.2.1 Overall schematic drawing

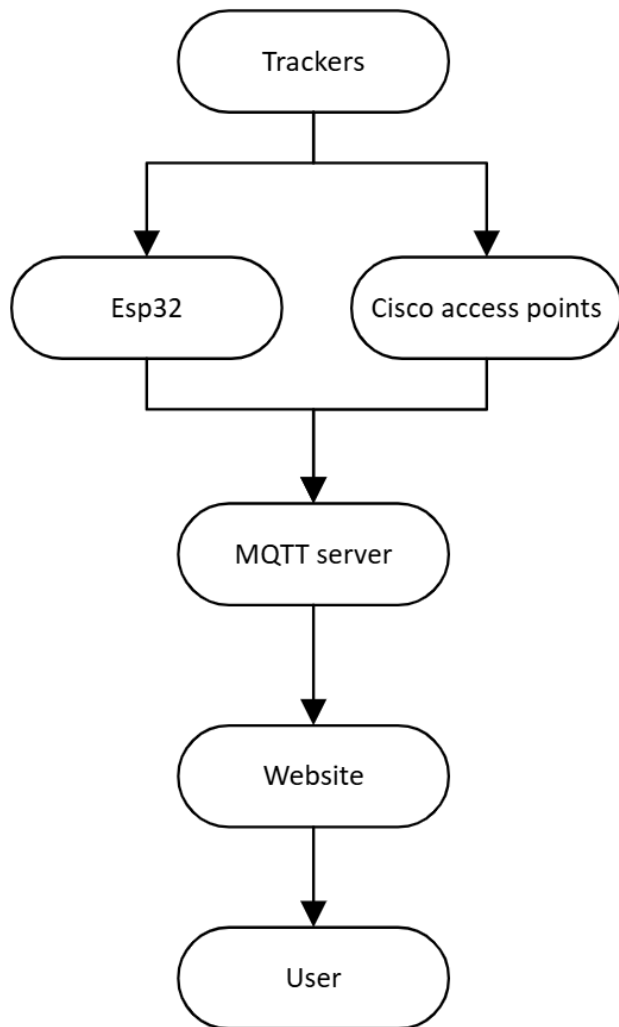


Figure 1: High level schematic drawing

1.2.2 Overall process diagram

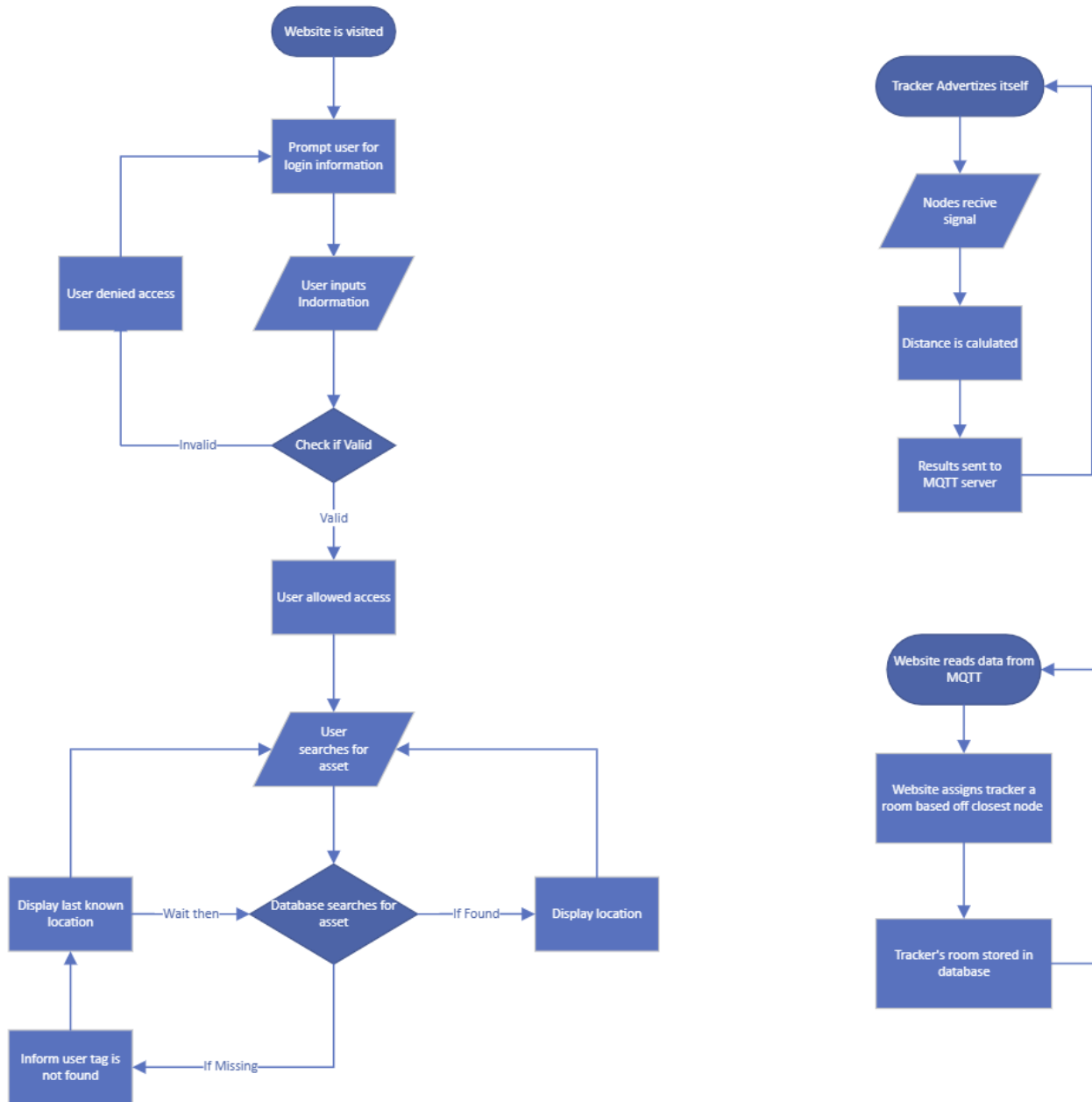


Figure 2: Process flow diagram for website front end, MQTT, and trackers

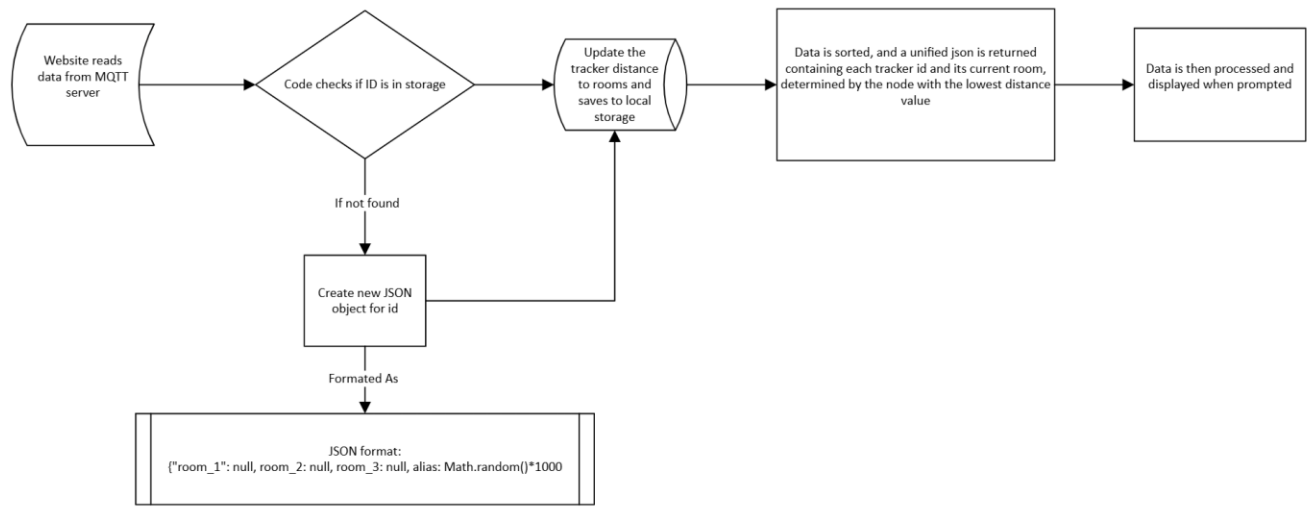


Figure 3: Website backend process flow diagram

2 Detail drawings

2.1 Part drawings

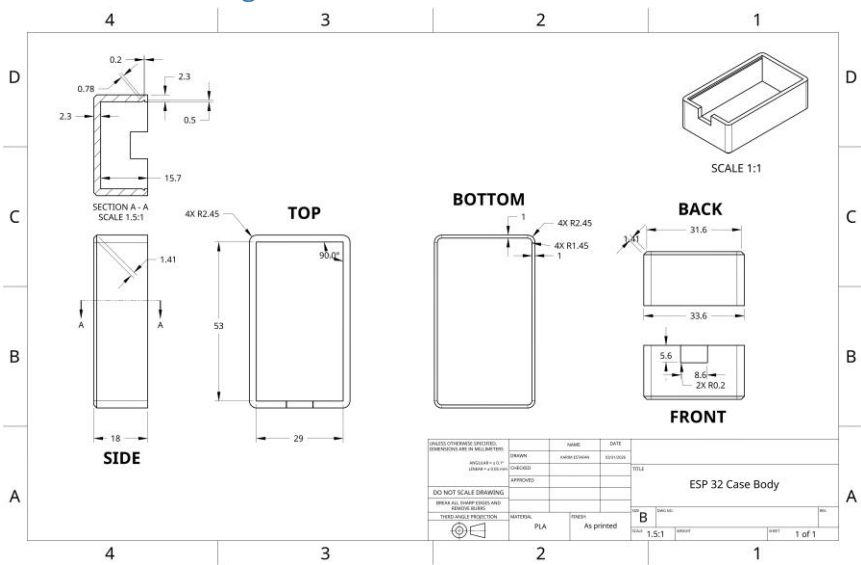


Figure 4: Esp32 case body drawing

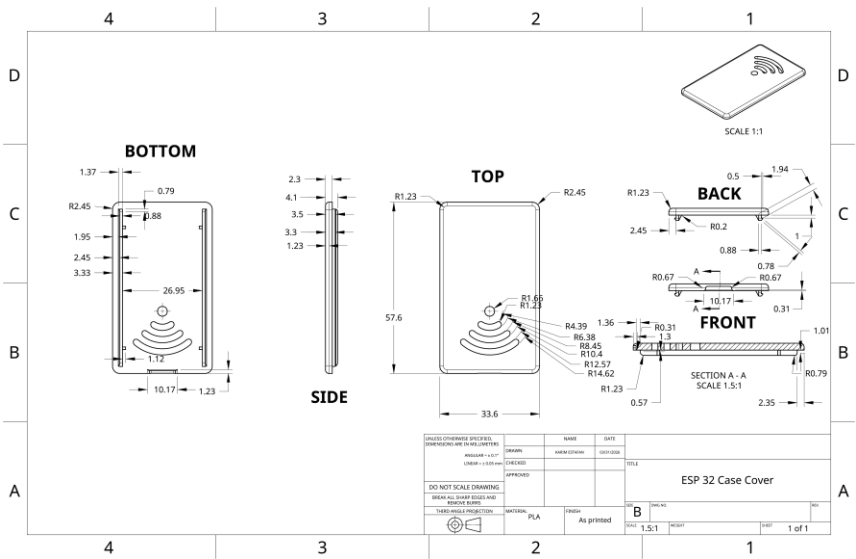


Figure 5: Esp32 case cover drawing

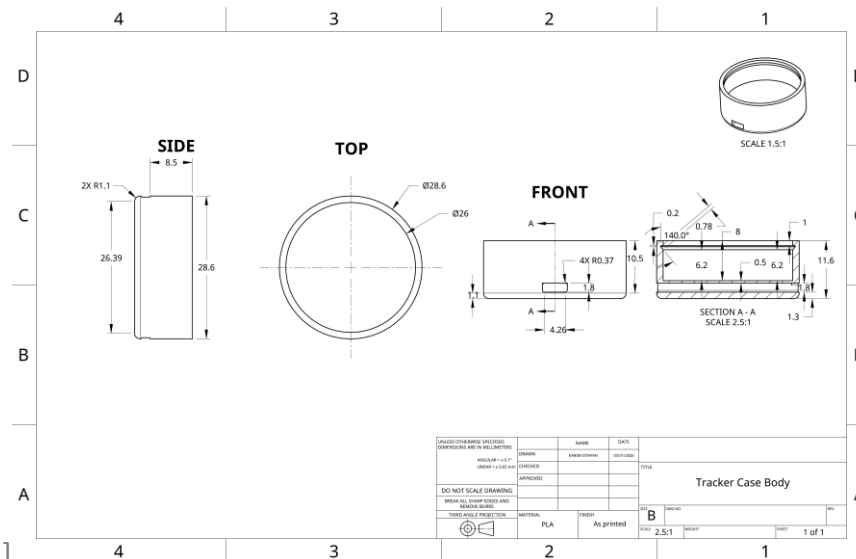


Figure 6: Tracker case body drawing

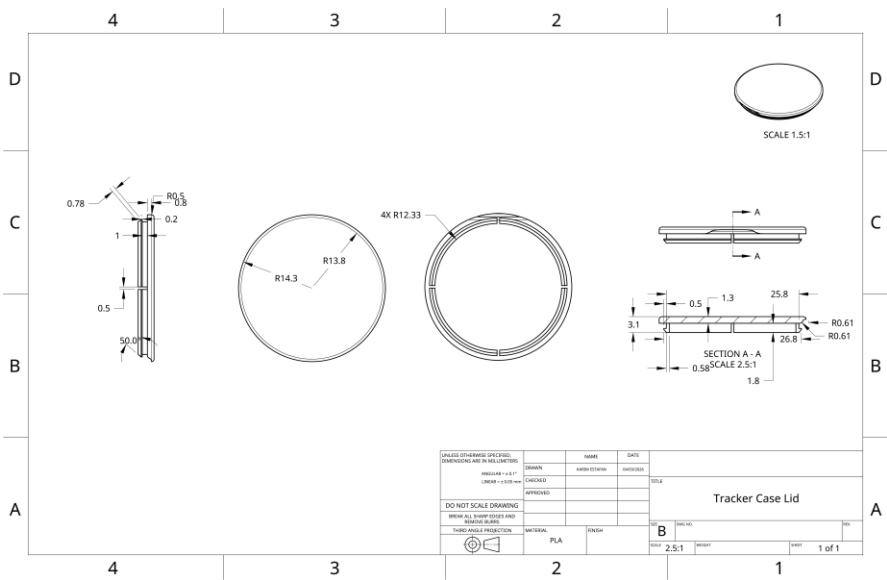


Figure 7: Tracker case lid drawing

2.2 Assembly drawings

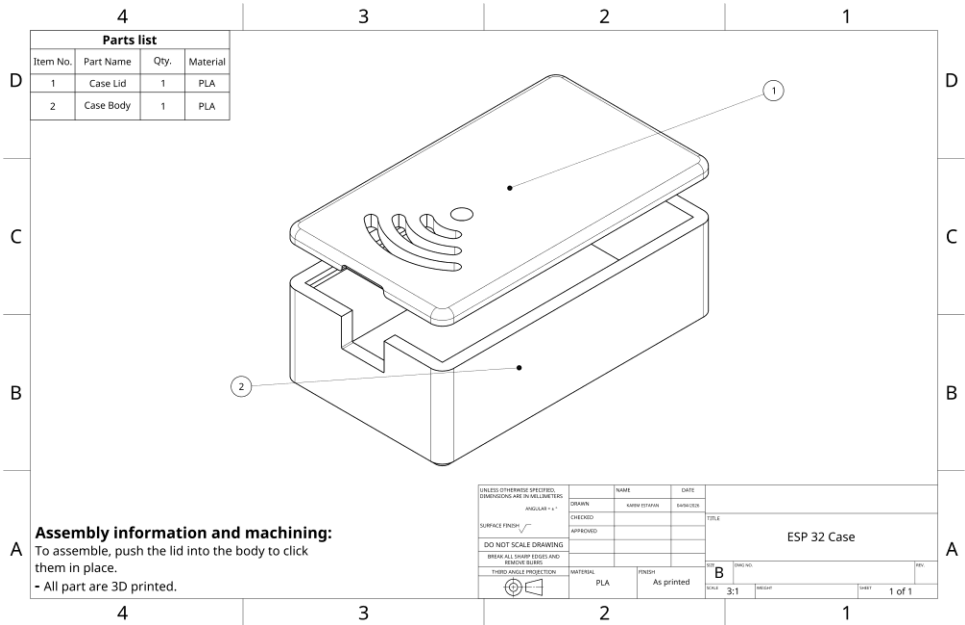


Figure 8: Esp32 case exploded view

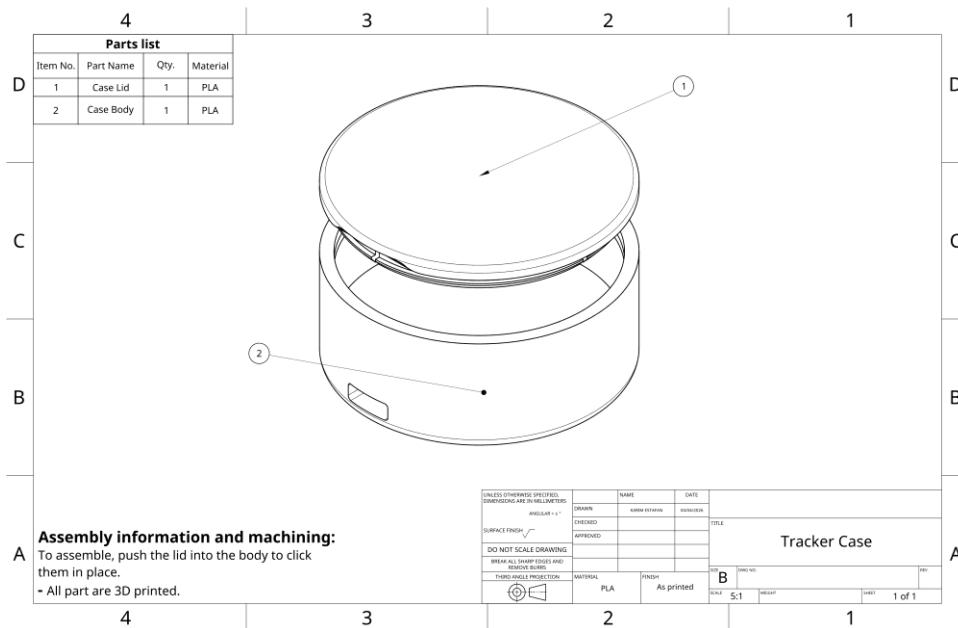


Figure 9: Tracker case exploded view

3 Software or Microelectronic Device Code

3.1 Code

The GitHub and information about setup can be found [here](#)

4 Digital User Interface

4.1 App or Webpage

Once you open the webpage you are presented with a floor plan and the available equipment. Clicking on an item will then increase the size of that asset on the map:

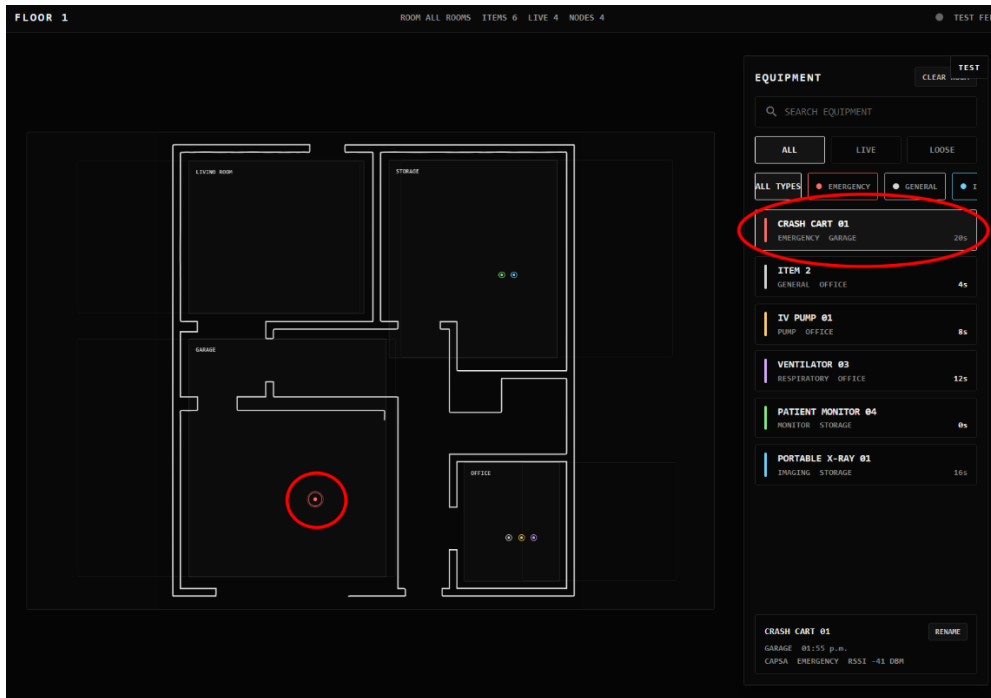


Figure 10: Home page of the website with crash cart highlighted

You can search for a specific asset, or for all the assets in a room using the search bar at the top of the page:

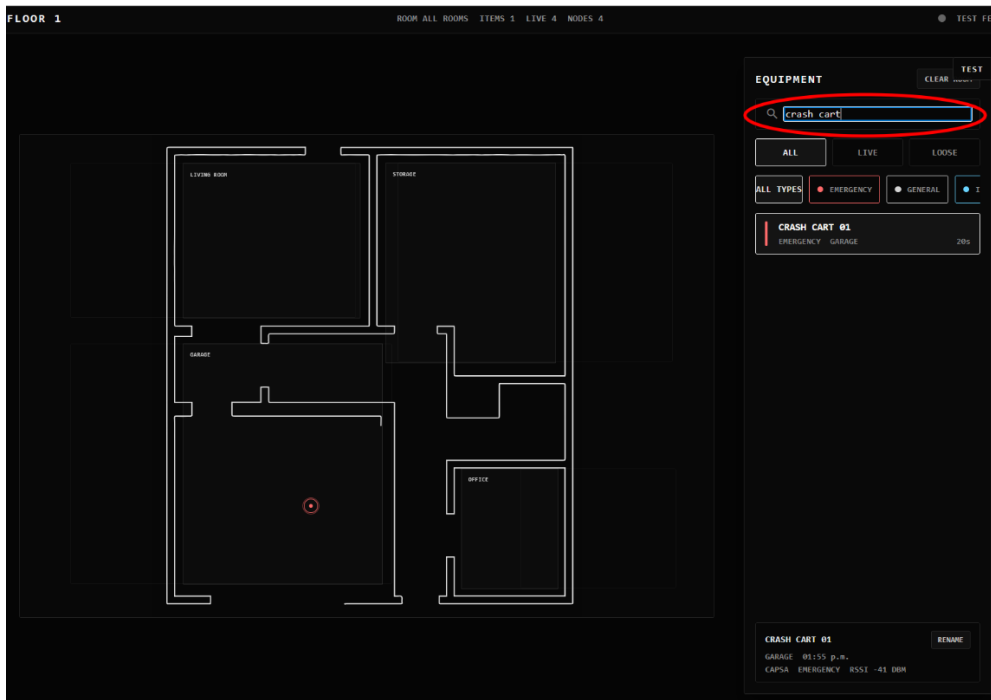


Figure 11: Search bar function of the website

You are also able to filter asset based on their type:

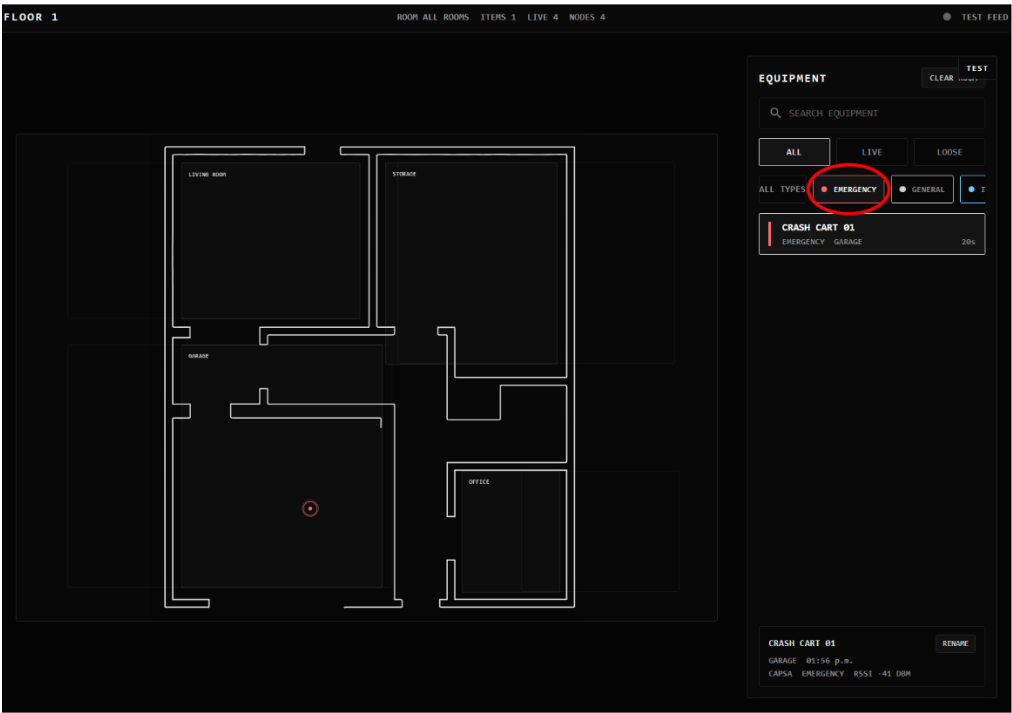


Figure 12: Filtering assets website functionality

To rename an asset right click on it then enter the preferred name in the field:

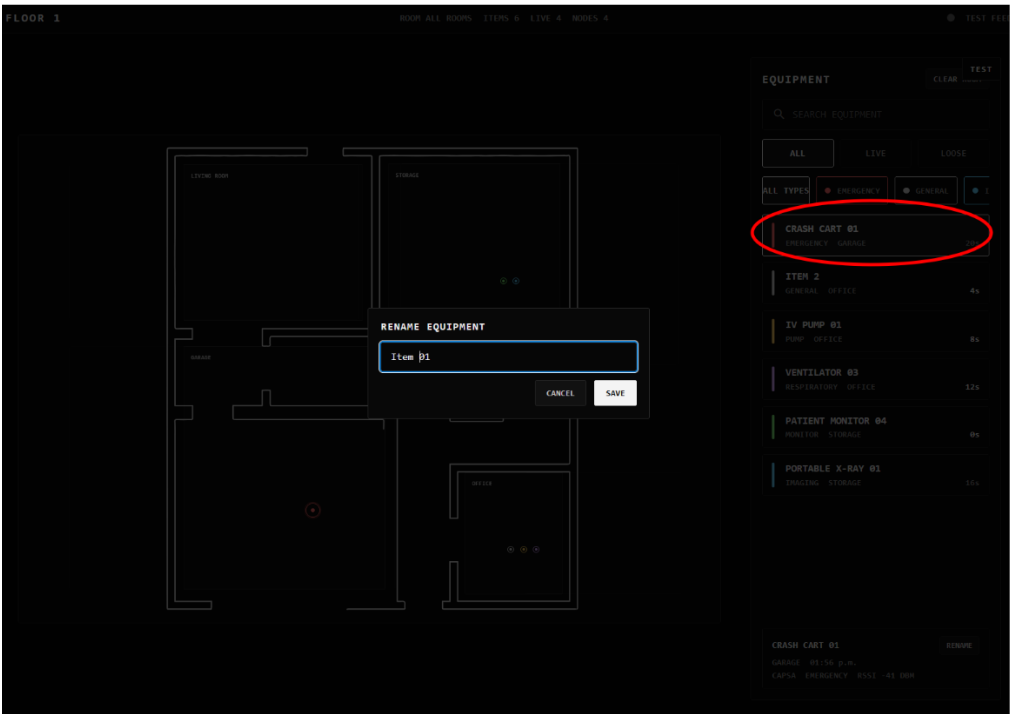


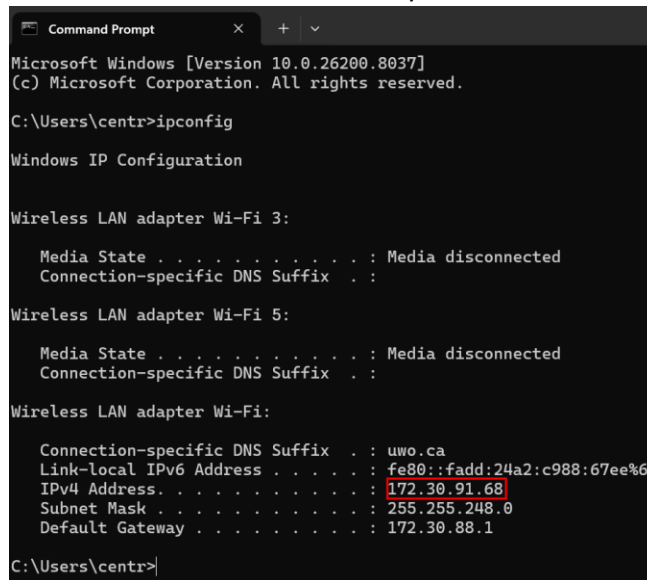
Figure 13: Website renaming function

5 Other

5.1 Fabrication, Assembly and Testing Instructions

Creation of the MQTT server:

1. Download broker from <https://mosquitto.org/> and run the setup executable
2. Open a text editor in administrator mode and navigate to 'mosquitto.conf'
 - a. At the bottom of the page paste:
 - i. listener 1883 0.0.0.0
 - protocol mqtt
 - listener 9001
 - protocol websockets
 - b. Click file then save
3. Open command prompt (Terminal) and enter 'ipconfig'
 - a. Note down the IPV4 address in my case it is 172.30.91.68



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Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\centr>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 5:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : uwo.ca
    Link-local IPv6 Address . . . . . : fe80::fadd:24a2:c988:67ee%6
    IPv4 Address. . . . . : 172.30.91.68
    Subnet Mask . . . . . : 255.255.248.0
    Default Gateway . . . . . : 172.30.88.1

C:\Users\centr>
  
```

Figure 14: Example searching for IPV4 address

- c.
4. Next move to the folder where Mosquitto is located on your computer
 - a. In general type 'cd "C:\Program Files\mosquitto"'
 - b. This may vary depending on your installation path
5. Next type '.\mosquitto_sub.exe -h YOUR IPV4 -v -t "#'
 - a. This is the ipv4 you just found
 - b. You should see a blank line, then data if information is being sent
 - c. To stop running press 'ctrl + c'
6. Now the server is running and listening for any data sent to it!

Esp 32 Node setup:

1. Plug in an ESP 32 to a computer
2. Navigate to Espresence download <https://espresense.com/firmware/>
3. Click connect and select the correct device
 - a. Add image
4. Click 'Download Firmware' and check erase device when prompted
5. Enter Wi-Fi data when prompted
 - a. If the connection times out just press the reset button on the Esp
6. Press 'Visit device'
7. Rename device to an appropriate name
8. Enter MQTT server address
 - a. This is just the IPV4 found above
9. Enter 1883 for port
10. Hit save
11. Now the Esp will scan for devices and report them to MQTT

Tracker setup:

1. Plug in an ESP 32 with Espresence configured
2. Navigate to device webpage using <https://espresense.com/firmware/>
3. Open footprints tab and sort by distance (closest on top)
4. Install battery into tracker and place it on top of the Esp
5. The top result should be the tracker
 - a. To ensure this is correct you can move the tracker away to see if it responds correctly
6. Note down the name of the device
 - a. This can be renamed in the website

3D Printing troubleshooting:

1. If the parts have trouble with warping a 5mm brim can be added to the exterior of the parts to help with adhesion. This will be removed after printed.
2. If tolerances are tight and the lids do not fit correctly you can remove 0.1mm off the lip of the lids to help get the right fit.

6 Bill of Materials

Tracker (1 Per asset to be tracked)							
#	Item Description	Units	Quantity	Unit cost	Cost	Source	Source 2
1	NRF52810 Tracking Beacon	1	1	\$15	\$ 15.00	Amazon	Aliexpress
2	Standard PLA filament (4.32g / Tracker)	1 KG	0.00432	\$24	\$ 0.10	Amazon 1	Amazon 2
3	Standard Ziptie	1000	0.001	\$14	\$ 0.01	Amazon	
				Total:	\$ 15.12		
Node (1 Node per room not covered by Cisco device)							
#	Item Description	Units	Quantity	Unit cost	Cost	Source	Source 2
1	ESP32 Devboard	1	1	\$10	\$ 10.00	Amazon	Aliexpress
2	Standard PLA filament (15.72g / Node)	1 KG	0.01572	\$24	\$ 0.38	Amazon 1	Amazon 2
				Total:	\$ 10.38		

Figure 15: Bill of materials

Source links:

Source	Source 2
Amazon	Aliexpress
Amazon 1	Amazon 2
Amazon	

Source	Source 2
Amazon	Aliexpress
Amazon 1	Amazon 2

* Note: - Any type of PLA filament will work

7 User Instructions (Operations and Maintenance)

Replacing tracker batteries

1. First open the case using the lip in the cap



a.

Figure 16: Tracker case with top removed

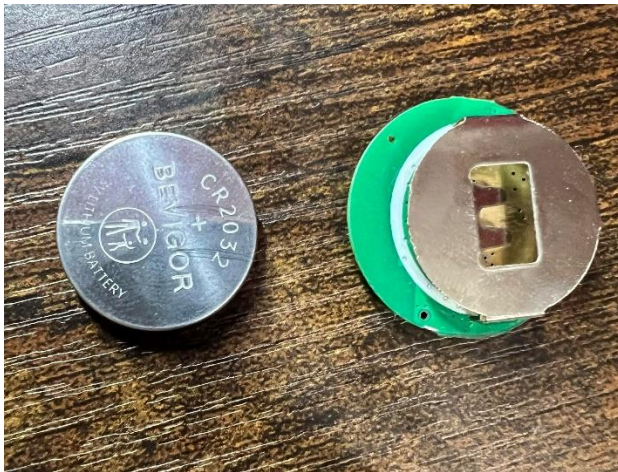
2. Then remove the tracker and use a nonconductive tool to push out the battery



a.

Figure 17: Removing battery out of the tracker

3. The replace with a new battery, positive side facing up towards the holder's prongs



a.

Figure 18: Placing battery into the tracker

8 Testing

Test Run	Purpose	Description of Test	Outcome	Insights Derived / Lessons Learned
			Time (s)	
1	Determine delay of tracker updates	Set up 1 Esp32 node and 1 tracker, set the tracker near Esp, then move it away. Measure how long it takes before the website shows the distance update	10	Tracker advertises itself on ~10 second intervals

Test Run	Purpose	Description of Test	Outcome	Insights Derived / Lessons Learned
			Distance	
1	Determine the accuracy of the tracker	Place node 1 meter away from tracker and measure distance reading	0.1m	Settings not calibrated correctly, RSSI showed -47 dBm at 1 meter
2	Determine the accuracy of the tracker	Updated RSSI calibration, placed tracker at 1 meter	0.78m	0.22m off
3	Determine the accuracy of the tracker	Placed tracker 2 meters away	3.13m	1.13m off
4	Determine the accuracy of the tracker	Placed tracker 5.3 meters away	5.52m	0.22m off

Test Run	Purpose	Description of Test	Outcome	Insights Derived / Lessons Learned
			Correct room	
1	Test the room sorting algorithm	Placed 2 trackers in 2 different rooms	FALSE	Did not account for delay between information being received by ESP's, caused issues where a tag would be compared with incorrect distances
2	Test the room sorting algorithm	Placed trackers in the same room	FALSE	Code didn't account for trackers in the same room as it was recording room first then looking for closest tracker to that room

3	Test the room sorting algorithm	Placed trackers in the same room then moved to a second room	TRUE	After fixing above issues trackers correctly appeared in rooms
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9 Validating compliance

9.1 Objectives validations

#	Objective	Assessment	Rationale / Evidence or Plan
1	Minimize cost	Met	\$15 per tracker, \$10 per node
2	Minimize latency	Exceeded	Average update time is 15s
3	Minimize size	Met	Tracker volume: 3.8 cm ³ , Node volume: 14.7 cm ³
4	Maximize operational life	Exceeded	Average power consumption is 10 μ A, Battery capacity 220 mAh therefore battery life is approximately ~2.5 years
5	Maximize ease of use	Partially met	Each asset must be manually configured, process takes around ~5 mins
6	Maximize repairability and replaceability	Met	Each asset works on its own and if 1 part fails only that part must be replaced
7	Maximize accuracy	Met	Tested at around 75% percent error

9.2 Constraints validations

#	Constraint	Assessment	Rationale / Evidence or Plan
1	Location inaccuracy no greater than 10m	Met	Testing displayed an accuracy within 1-2 meters
2	Tracking latency, no greater than 5 min	Met	Testing displayed a 10-15 second update delay
3	Use existing infrastructure	Met	Our solution is compatible with existing cisco access points
4	Withstands against cleaning agents	Not verified yet	Submerge case into cleaning solution and ensure there is no material deterioration
5	Implementable at a large scale (100's of assets)	Met	Each asset would have its own tracker and would just be added to the website using the add asset function
6	Meets Canadian health standards	Not verified yet	Consult accredited 3rd party organization
7	Reliable operation across multiple floors	Met	Testing showed consistently accurate readings across floors
8	Lifespan of ~5 yrs	Not verified yet	Accelerate wear and tear by increasing update frequency, and test durability by dropping devices

9	Greater than 6 months of battery life	Met	Calculated from average power consumption and battery capacity
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